



2011 Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 11th May 2011

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- All necessary working should be shown in every question.
- Start each question on a new leaflet.

Structure of the paper

- Total marks — 96
- All eight questions may be attempted.
- All eight questions are of equal value.

Collection

- Write your name, class and master clearly on each leaflet.
- Hand in the eight questions in a single well-ordered pile.
- Hand in a leaflet for each question, even if it has not been attempted.
- If you use a second leaflet for a question, place it inside the first.
- Write your name on the question paper and place it inside your leaflet for Question One.

5A: BDD
5D: KWM
5G: MW

5B: PKH
5E: SJE
5H: SO

5C: FMW
5F: RCF
5I: MLS

Checklist

- Writing leaflets: 8 per boy.
- Candidature — 149 boys

Examiner
RCF

QUESTION ONE (12 marks) Start a new leaflet.

(a) Simplify:

(i) $\sqrt{48} - \sqrt{27}$

(ii) $(2\sqrt{5} + 5)^2$

(b) (i) Write down the exact value of $\cos 45^\circ$.

(ii) Find the exact value of $\sec 135^\circ$.

(c) Differentiate the following functions:

(i) $2x^3 - x^5$

(ii) x^{-3}

(iii) $\sqrt[3]{x}$

(d) Find the equation of the straight line perpendicular to the line $y = 3x - 5$ and passing through the point $(-1, 2)$. Give your answer in general form.

(e) Factorise $x^3 + 27$.

QUESTION TWO (12 marks) Start a new leaflet.

(a) Write down the domain of the function $y = \frac{1}{x-3}$.

(b) Consider the sequence $2, \frac{2}{3}, \frac{2}{9}, \dots$

(i) Show the sequence is geometric.

(ii) State the values of a and r .

(iii) Find the limiting sum of the geometric series $2 + \frac{2}{3} + \frac{2}{9} + \dots$.

(c) Given the points $A(-2, 5)$ and $B(1, -1)$, find the length of the interval AB . Give your answer in simplest exact form.

(d) Solve $|x - 4| = 5$.

(e) The area of $\triangle ABC$ is 14 cm^2 . Given that $AB = 8 \text{ cm}$ and $AC = 7 \text{ cm}$, find all possible values of $\angle BAC$.

(f) Solve $(x + 3)(x - 4) \geq 0$.

QUESTION THREE (12 marks) Start a new leaflet.

- (a) By rationalising the denominator and simplifying, find the values of a , b and c such that

$$\frac{3 + \sqrt{12}}{2 - \sqrt{3}} = a + b\sqrt{c}.$$

- (b) Given that $\sin \theta = \frac{5}{7}$ and θ is obtuse, find in simplest form the exact values of:

(i) $\tan \theta$

(ii) $\sec \theta$

- (c) Complete the square to find the centre and the radius of the circle with equation

$$x^2 + y^2 + 10y = 39.$$

- (d) Show that $\frac{\cos^2 x - \sin^2 x}{\cos^2 x + \cos x \sin x} = 1 - \tan x$.

- (e) Find the perpendicular distance from the point $(4, 2)$ to the line $3x + 4y - 5 = 0$.

QUESTION FOUR (12 marks) Start a new leaflet.

- (a) Given $f(x) = \sqrt{x}(3x^2 - 2x + 1)$, find $f'(x)$.

- (b) Given the points $A(-2, 5)$ and $B(7, -1)$, find the point P which divides the interval AB internally in the ratio $2 : 1$.

- (c) (i) If $f(x) = x^2 + 3x$, find $f(x + h)$.

- (ii) Hence differentiate $f(x)$ from first principles.

- (d) The third term of an AP is 125 and the sixth term is 116.

- (i) Find the first term a and the common difference d .

- (ii) Find the value of the thirtieth term.

- (iii) Find the minimum number of terms to be added for the sum to be negative.

QUESTION FIVE (12 marks) Start a new leaflet.

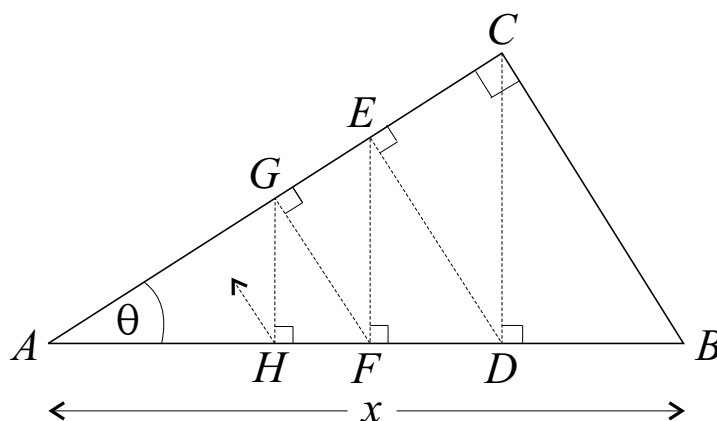
- (a) (i) Sketch the graph of $y = \tan x$ for the restricted domain $0^\circ \leq x \leq 360^\circ$.
Clearly indicate all intercepts and asymptotes.
- (ii) Using your graph and knowledge of special angles, solve $\tan x = -\sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$.
- (b) Use the chain, product or quotient rules to differentiate the following functions:
- (i) $\frac{5x+1}{5-x}$
- (ii) $(x^3+3)(3x-2)^3$ (Give your final answer in factored form.)
- (c) The lines $x-2y+3=0$ and $3x+y-2=0$ intersect at point A . Without finding the co-ordinates of A , find the equation of the line that passes through A and $(2,1)$.
Give your answer in general form.

QUESTION SIX (12 marks) Start a new leaflet.

- (a) Find the values of x for which the geometric series $(2-x) + (2-x)^2 + (2-x)^3 + \dots$ has a limiting sum.
- (b) Solve the following equations over the domain $0^\circ \leq \theta \leq 360^\circ$.
- (i) $\operatorname{cosec} \theta = -\sqrt{2}$
- (ii) $2 \sin^2 \theta + \cos \theta = 1$
- (iii) $\cos(2\theta + 60^\circ) = \frac{1}{\sqrt{2}}$

QUESTION SEVEN (12 marks) Start a new leaflet.

- (a) Sketch the graph of the function $y = 2^{-x} - 4$, clearly indicating any asymptotes and intercepts with the axes.
- (b) Solve $\frac{3x}{x-2} \geq 1$.
- (c)



In the diagram above, let side $AB = x$ and $\angle BAC = \theta$

- Write down an expression for the side BC .
- Find expressions for CD and EF in terms of x and θ .
- Hence find the limiting sum of the series $CD + EF + GH + \dots$.

QUESTION EIGHT (12 marks) Start a new leaflet.

- (a) Consider the points $A(2, 1)$ and $B(-2, 4)$ in the number plane.
- Write down the equation of a line with gradient m that passes through A . Give your answer in general form.
 - Find the equations, in general form, of the two lines ℓ_1 and ℓ_2 that pass through A and are 3 units from B .
 - Find the equations of the two lines that bisect the angles at A between ℓ_1 and ℓ_2 .
- (b) Consider the function $f(x) = \frac{x+4}{2x^2-8}$.
- Determine whether $f(x)$ is even, odd or neither.
 - Find all intercepts and asymptotes of the graph of $y = f(x)$.
 - Find the gradient function $f'(x)$, and hence find the x co-ordinates of the two points where the graph has zero gradient.
 - Sketch the graph of the function $y = f(x)$.

END OF EXAMINATION

HY 2011 - FIFTH FORM EXTENSION ONE (RCF)

Qn 1

$$\begin{aligned} \text{a) (i)} \quad \sqrt{48} - \sqrt{27} \\ = 4\sqrt{3} - 3\sqrt{3} \\ = \sqrt{3} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad (2\sqrt{5} + 5)^2 &= (2\sqrt{5} + 5)(2\sqrt{5} + 5) \\ &= 20 + 10\sqrt{5} + 10\sqrt{5} + 25 \\ &= 45 + 20\sqrt{5} \quad \checkmark \\ &\{ \text{or } 5(9 + 4\sqrt{5}) \} \end{aligned}$$

$$\text{b) (i)} \quad \cos 45^\circ = \frac{1}{\sqrt{2}} \quad \{ \text{or } \frac{\sqrt{2}}{2} \} \quad \checkmark$$

$$\text{(ii)} \quad \sec 135^\circ = \frac{1}{\cos 135^\circ} = \frac{1}{-\frac{1}{\sqrt{2}}} = -\sqrt{2} \quad \checkmark$$

$$\begin{aligned} \text{c) (i)} \quad y &= 2x^3 - 5x^5 \\ \frac{dy}{dx} &= 6x^2 - 25x^4 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad y &= x^{-3} \\ \frac{dy}{dx} &= -3x^{-4} \quad \checkmark \\ &\{ \text{or } -\frac{3}{x^4} \} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad y &= x^{\frac{1}{3}} \quad \checkmark \\ \frac{dy}{dx} &= \frac{1}{3}x^{-\frac{2}{3}} \quad \checkmark \\ &\{ \text{or } \frac{1}{3\sqrt[3]{x^2}} \} \end{aligned}$$

$$\text{d) } y = 3x - 5 \quad m_1 = 3 \quad \therefore \text{Perp Grad } m_2 = \left(-\frac{1}{3}\right) \quad \text{thru } (-1, 2)$$

$$y - 2 = -\frac{1}{3}(x - (-1)) \quad \checkmark$$

$$3y - 6 = -x - 1$$

$$x + 3y - 5 = 0 \quad \checkmark$$

$$\text{e) } x^3 + 27 = (x + 3)(x^2 - 3x + 9) \quad \checkmark$$

$$\frac{1}{12}$$

Qn 2

a) $x-3 \neq 0$
 $\therefore x \in \mathbb{R}, x \neq 3 \checkmark$

b) $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}$

$\frac{t_3}{t_2} = \frac{\frac{2}{9}}{\frac{2}{3}} = \frac{1}{3}$ $\frac{t_2}{t_1} = \frac{\frac{2}{3}}{2} = \frac{1}{3} \checkmark$

$\therefore$ GP common ratio $r = \frac{1}{3}$

(ii) $a = 2, r = \frac{1}{3} \checkmark$

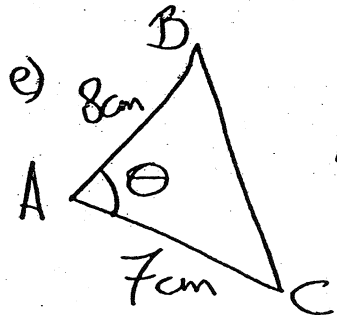
(iii) $S_{\infty} = \frac{a}{1-r} = \frac{2}{1-\frac{1}{3}} = \frac{2}{\frac{2}{3}} = 3 \checkmark$, since $|r| < 1$

c) A (-2, 5) B (1, -1)

$AB = \sqrt{(5-(-1))^2 + (-2-1)^2} \checkmark$
 $= \sqrt{6^2 + 3^2}$
 $= \sqrt{36+9} = \sqrt{45} = 3\sqrt{5} \text{ u} \checkmark$

d) $|x-4| = 5$

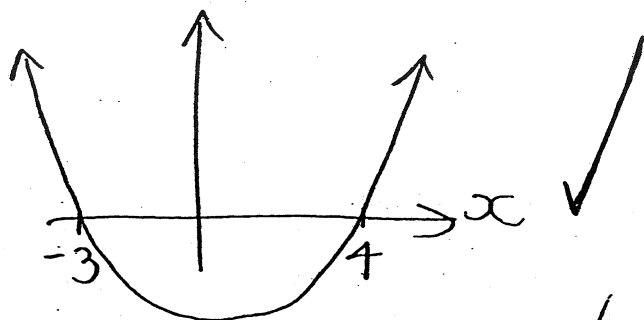
i.e. $x-4 = 5$ / or $x-4 = -5$
 $x = 9 \checkmark$ or $x = -1 \checkmark$



Area = 14 cm^2

$A_{\Delta} = \frac{1}{2} bc \sin A$
 $\therefore 14 = \frac{1}{2} \times 7 \times 8 \times \sin \theta \checkmark$
 $\frac{1}{2} = \sin \theta$
 $\therefore \theta = 30^\circ \text{ or } 150^\circ \checkmark$

e) $(x+3)(x-4) \geq 0$



$\therefore x \geq 4, \text{ or } x \leq -3 \checkmark$

(12)

Q11

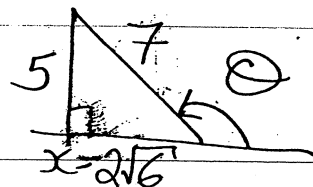
$$a) \frac{3+\sqrt{12}}{2-\sqrt{3}} = \frac{(3+\sqrt{12})(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} \checkmark$$

$$= \frac{6+2\sqrt{12}+3\sqrt{3}+6}{4-3}$$

$$= 12+7\sqrt{3} \checkmark - a+b\sqrt{c}$$

$$\therefore a=12, b=7, c=3 \checkmark$$

$$b) \sin \theta = \frac{5}{7}$$



$$\therefore x = \sqrt{49-25}$$

$$= \sqrt{24}$$

$$= 2\sqrt{6} \checkmark$$

obtuse $\Rightarrow$ 2nd quad $\checkmark$

$$(i) \tan \theta = -\frac{5}{2\sqrt{6}} \checkmark$$

$$(ii) \sec \theta = \frac{1}{\cos \theta} = \frac{-7}{2\sqrt{6}} = \left(-\frac{7\sqrt{6}}{12}\right)$$

$$c) x^2 + y^2 + 10y = 39$$

$$x^2 + y^2 + 10y + 25 = 64$$

$$x^2 + (y+5)^2 = 64$$

Circle: centre (0, -5)
radius 8u

$$d) \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \cos x \sin x} = \frac{(\cos x + \sin x)(\cos x - \sin x)}{\cos x (\cos x + \sin x)} \checkmark$$

$$= \frac{\cos x - \sin x}{\cos x} \checkmark$$

$$= 1 - \tan x$$

$$= \text{RHS} \blacksquare$$

$$e) (4, 2) \quad 3x + 4y - 5 = 0$$

$$P = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3 \times 4 + 4 \times 2 - 5|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|12 + 8 - 5|}{5}$$

$$= 3 \text{ units}$$

12

Qu 4

a) $y = \sqrt{x} (3x^2 - 2x + 1)$

$$y = 3x^{5/2} - 2x^{3/2} + x^{1/2} \checkmark$$

$$\frac{dy}{dx} = \frac{15}{2}x^{3/2} - 3x^{1/2} + \frac{1}{2}x^{-1/2} \checkmark$$

$$\left\{ \text{or } \frac{15(\sqrt{x})^3 - 6\sqrt{x} + \frac{1}{\sqrt{x}}}{2} \right\}$$

$$\left\{ \text{or } \frac{15x^2 - 6x + 1}{2\sqrt{x}} \right\}$$

b) A(-2, 5) B(7, -1) $k:l = 2:1$

$$P: x_p = \frac{l x_A + k x_B}{k+l}$$

$$= \frac{1 \times (-2) + 2 \times 7}{2+1}$$

$$= \frac{12}{3}$$

$$= 4 \checkmark$$

$$y_p = \frac{l y_A + k y_B}{2+1}$$

$$= \frac{-5}{3} \checkmark$$

$$\therefore P(4, -\frac{5}{3})$$

c) (i) $f(x) = x^2 + 3x$

$$f(x+h) = (x+h)^2 + 3(x+h) \checkmark$$

$$= x^2 + 2xh + h^2 + 3x + 3h$$

$$(ii) f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{(x+h) - x} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{2xh + h^2 + 3h}{h} \right) \checkmark$$

$$= \lim_{h \rightarrow 0} (2x + 3 + h)$$

$$= 2x + 3$$

$$\begin{aligned} \text{d)(i) AP } t_3 &= a + 2d = 125 \text{ ①} \\ t_6 &= a + 5d = 116 \text{ ②} \end{aligned} \quad \left. \vphantom{\begin{aligned} t_3 &= a + 2d = 125 \text{ ①} \\ t_6 &= a + 5d = 116 \text{ ②} \end{aligned}} \right\} \checkmark$$

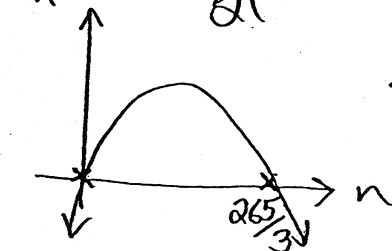
$$\begin{aligned} \text{②} - \text{①} &: 3d = -9 \\ & d = (-3) \end{aligned} \quad \left. \vphantom{\begin{aligned} 3d &= -9 \\ d &= (-3) \end{aligned}} \right\} \checkmark \begin{aligned} \text{sub into ① } & a - 6 = 125 \\ & a = 131 \end{aligned} \quad \therefore \text{First term, } a = 131 \\ & \text{Common difference, } d = (-3)$$

$$\begin{aligned} \text{(ii) } t_{30} &= a + 29d \\ &= 131 + 29(-3) \\ &= 131 - 87 \\ &= 44 \end{aligned} \quad \checkmark$$

$$\begin{aligned} \text{(iii) } S_n &= \frac{n}{2}(2a + (n-1)d) \\ S_n &= \frac{n}{2}(262 + (n-1)(-3)) \\ &= \frac{n}{2}(265 - 3n) \end{aligned} \quad \checkmark$$

Require $S_n < 0$

$$\therefore \frac{n}{2}(265 - 3n) < 0 \quad \checkmark$$

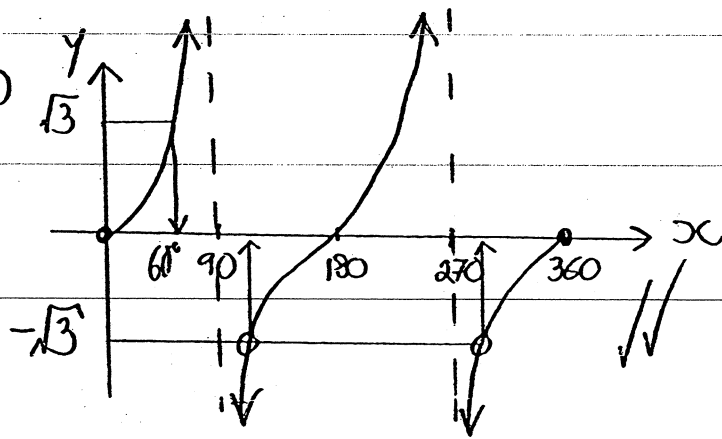


since $n > 0$
 $n \in \mathbb{Z}$

$$\begin{aligned} n &> \frac{265}{3} \\ n &= 89 \quad \checkmark \text{ need } S_{89} \end{aligned}$$

12

Qn 5 a) (i)



(ii) $\tan x = (-\sqrt{3})$

$\Rightarrow x = 90 + 30$
 $= 120^\circ$ ✓

OR $x = 270 + 30$
 $= 300^\circ$ ✓

b) (i) $y = \frac{5x+1}{5-x}$

Quotient Rule: Let $u = 5x+1$ $v = 5-x$

$\frac{dy}{dx} = \frac{(5-x) \times 5 - (5x+1)(-1)}{(5-x)^2}$ ✓

$\frac{du}{dx} = 5$

$\frac{dv}{dx} = (-1)$

$= \frac{25 - 5x + 5x + 1}{(5-x)^2}$ ✓

$= \frac{26}{(5-x)^2}$ ✓

(ii) $y = (x^3+3)(3x-2)^3$ Product Rule: Let $u = (x^3+3)$, $v = (3x-2)^3$

$\frac{dy}{dx} = (3x-2)^3 \times 3x^2 + (x^3+3) \times 9(3x-2)^2$ ✓

$\frac{du}{dx} = 3x^2$ $\frac{dv}{dx} = 3(3x-2)^2$
 $= 9(3x-2)^2$

$= 3(3x-2)^2 [x^2(3x-2) + 3(x^3+3)]$

$= 3(3x-2)^2 [3x^3 - 2x^2 + 3x^3 + 9] = 3(3x-2)^2 (6x^3 - 2x^2 + 9)$ ✓

c) $l_1: x - 2y + 3 = 0$

$l_2: 3x + y - 2 = 0$

Concurrent lines $x - 2y + 3 + k(3x + y - 2) = 0$ ✓

$(1+3k)x + (k-2)y + (3-2k) = 0$

Passes thru (2,1) $\therefore 2(1+3k) + (k-2) + (3-2k) = 0$

$2 + 6k + k - 2 + 3 - 2k = 0$

$5k + 3 = 0$

$k = (-\frac{3}{5})$ ✓

$\therefore$ Line is $-\frac{4}{5}x - \frac{13}{5}y + \frac{21}{5} = 0$ ✓

$4x + 13y - 21 = 0$ ✓

(12)

Qn 6

a) $(2-x) + (2-x)^2 + (2-x)^3 + \dots$

GP $a = (2-x)$
 $r = (2-x)$

for limiting sum $|r| < 1$

$-1 < 2-x < 1$ ✓

$-3 < -x < -1$ ✓

$3 > x > 1$ ✓

ie $1 < x < 3$

b) (i) $\cos \theta = -\sqrt{2}$

$\sin \theta = -\frac{1}{\sqrt{2}}$ ✓ $\therefore 3^{\text{rd}} \& 4^{\text{th}}$ quadrants

$\theta = 225^\circ, 315^\circ$ ✓

(ii) $2\sin^2 \theta + \cos \theta = 1$

$2(1 - \cos^2 \theta) + \cos \theta - 1 = 0$ ✓

$1 - 2\cos^2 \theta + \cos \theta = 0$

$2\cos^2 \theta - \cos \theta - 1 = 0$ ✓

$(2\cos \theta + 1)(\cos \theta - 1) = 0$ ✓

$\cos \theta = (-\frac{1}{2})$ or 1

$\theta = 120^\circ, 240^\circ, 0^\circ, 360^\circ$ ✓

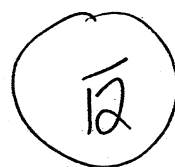
(iii) $\cos(2\theta + 60^\circ) = \frac{1}{\sqrt{2}}$

Change Domain

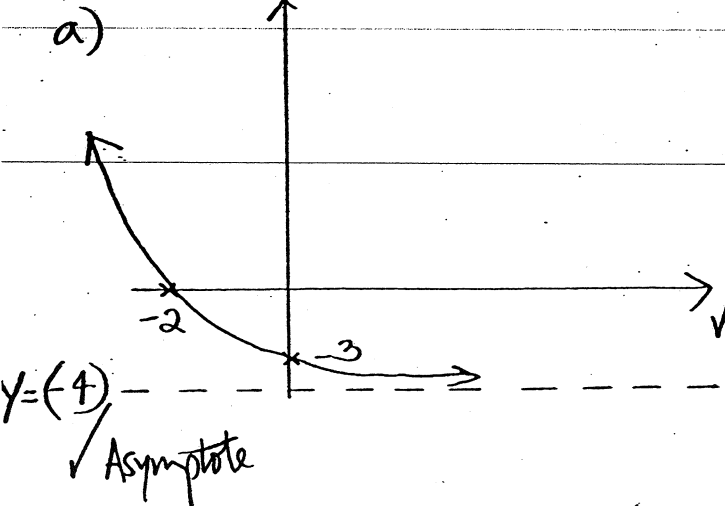
$2\theta + 60^\circ = 315^\circ, 405^\circ, 675^\circ, 765^\circ$

$2\theta = 255^\circ, 345^\circ, 615^\circ, 705^\circ$

$\theta = 127\frac{1}{2}^\circ, 172\frac{1}{2}^\circ, 307\frac{1}{2}^\circ, 352\frac{1}{2}^\circ$ ✓



Qn 7

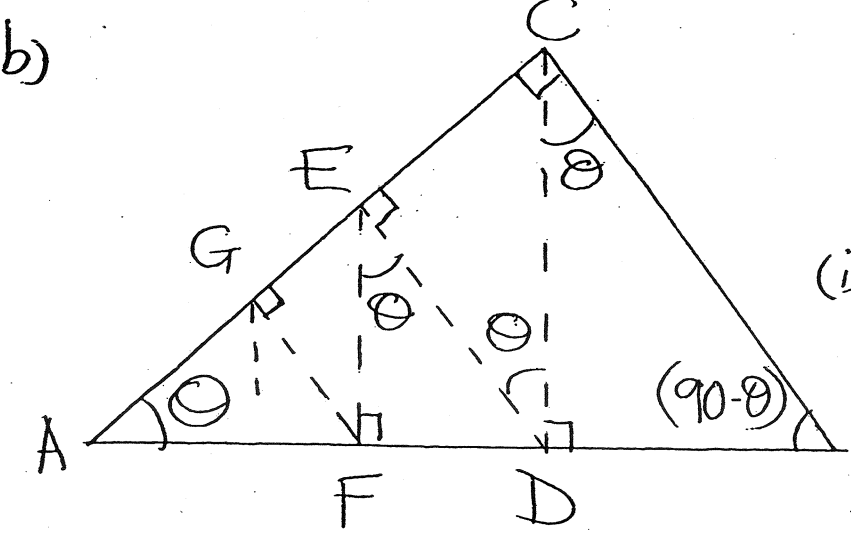
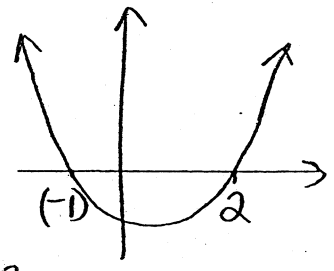


$y = 2^x$
 Reflect in y axis
 $y = 2^{-x}$
 Translate down 4u
 $y = 2^{-x} - 4$

$x = 0$
 $y = 2^{-0} - 4$
 $= 1 - 4$
 $= -3$
 $(0, -3)$
 $y = 0$
 $2^{-x} = 4$
 $\frac{1}{2^x} = 4$ Both
 $\therefore 2^x = \frac{1}{4}$ Into
 $x = -2$
 $(-2, 0)$ ✓

b) $\frac{3x}{x-2} \geq 1$ $\times (x-2)^2$ ✓ Dom: $x \neq 2$

$3x(x-2) \geq (x-2)^2$
 $3x(x-2) - (x-2)^2 \geq 0$
 $(x-2)[3x - (x-2)] \geq 0$ ✓
 $(x-2)(2x+2) \geq 0$ ✓
 $(x-2)(x+1) \geq 0$
 $x \leq -1$ / or $x \geq 2$ ✓ (NB $x \neq 2$)



$\angle CBA = 90 - \theta$ (Angle Sum of $\triangle ABC$)

(i) in $\triangle ABC$
 $\sin \theta = \frac{BC}{AB}$
 $BC = x \sin \theta$ ✓

(ii) in $\triangle CBD$
 $\sin(90 - \theta) = \frac{CD}{BC}$
 $(x \sin \theta) \cos \theta = CD$ ✓
 $\angle BCD = \theta$ (Angle Sum of $\triangle BCD$)

in $\triangle DEC$ $\angle CDE = \theta$ (Alternate angles $BC \parallel DE$)

$$\therefore \cos \theta = \frac{DE}{CD}$$

$$DE = x \sin \theta \cos^2 \theta$$

in $\triangle DEF$ $\angle FED = \theta$ (Alternate Angles $CD \parallel EF$)

$$\therefore \cos \theta = \frac{EF}{DE}$$

$$EF = x \sin \theta \cos^3 \theta \quad \checkmark$$

(iii) $CD + EF + GH + \dots$

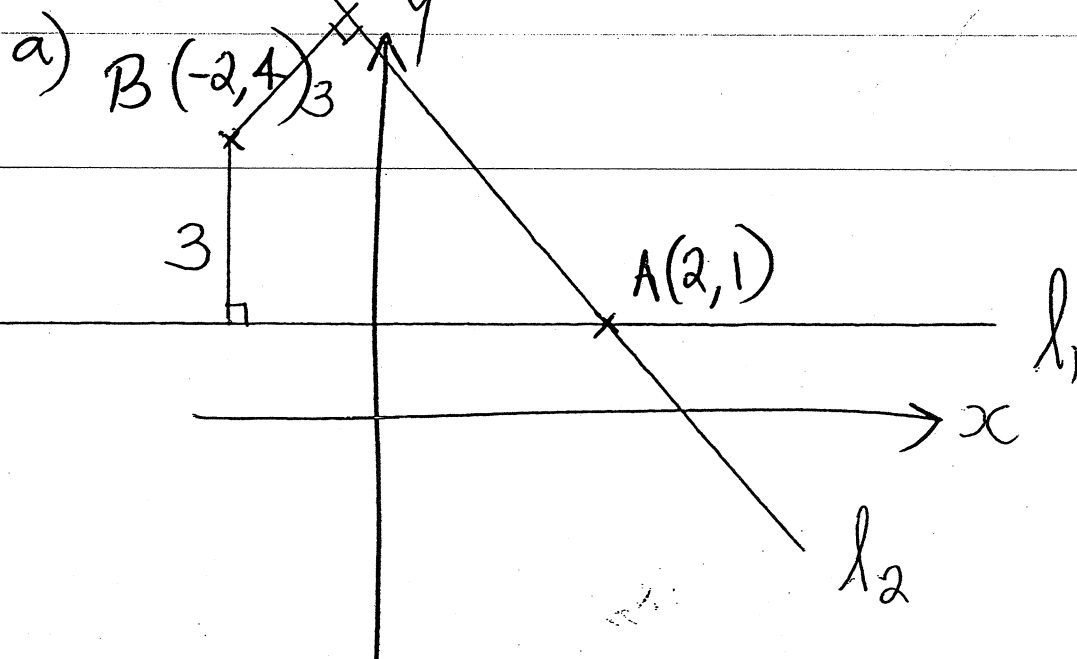
$$x \sin \theta \cos \theta + x \sin \theta \cos^3 \theta + x \sin \theta \cos^5 \theta + \dots$$

GP $a = x \sin \theta \cos \theta$ $\checkmark$
 $r = \cos^2 \theta$

$$\begin{aligned} S_{\infty} &= \frac{a}{1-r} = \frac{x \sin \theta \cos \theta}{1 - \cos^2 \theta} \\ &= \frac{x \sin \theta \cos \theta}{\sin^2 \theta} \\ &= \underline{x \cot \theta} \quad \checkmark \end{aligned}$$

(12)

Qn 8



(i) $y-1 = m(x-2)$
 $y-1 = mx-2m$
 $0 = mx-y+(1-2m)$ ✓ (or Gradient-Intercept accepted)

(ii) Perp distance from $(-2,4)$ to $mx-y+(1-2m)$ is 3

$$\frac{|-2m-4+1-2m|}{\sqrt{m^2+(-1)^2}} = 3 \quad \checkmark$$

$$|-3-4m| = 3\sqrt{m^2+1}$$

(sq) $(3+4m)^2 = 9m^2+9$

$$9+24m+16m^2 = 9m^2+9$$

$$24m+7m^2 = 0$$

$$m(24+7m) = 0$$

$$m=0 \text{ or } m = \left(-\frac{24}{7}\right) \quad \checkmark$$

$l_1: y=1$ $l_2: 0 = -\frac{24}{7}x - y + \left(1 + \frac{48}{7}\right)$

$$0 = -24x - 7y + 55$$

$$24x + 7y - 55 = 0 \quad \checkmark \text{ Both}$$

One angle bisector is line AB.

$$m_{AB} = \frac{4-1}{-2-2} = \left(-\frac{3}{4}\right)$$

Eqn AB $y-1 = -\frac{3}{4}(x-2)$

$$4y-4 = -3x+6$$

$$3x+4y-10=0$$

Perp Line $m_2 = \frac{4}{3}$

Eqn other angle bisector

$$y-1 = \frac{4}{3}(x-2)$$

$$3y-3 = 4x-8$$

$$4x-3y-5=0$$

Angle bisectors are $3x+4y-10=0$ ✓
and $4x-3y-5=0$.

i) $f(x) = \frac{x+4}{2x^2-8}$
 $= \frac{x+4}{2(x+2)(x-2)}$

Dom: $x \neq 2$ or (-2)

(i) $f(-x) = \frac{-x+4}{2(-x)^2-8}$

$$= \frac{-x+4}{2x^2-8} \neq f(x)$$

$\neq -f(x) \therefore$ No Symmetry ✓

(ii) y-intercept $x=0$

x-intercept $y=0$

Vertical Asymptotes

Horizontal Asymptote

$$f(0) = \frac{4}{-8} = \left(-\frac{1}{2}\right) \quad (0, -\frac{1}{2})$$

$$x+4=0 \quad x=(-4) \quad (-4, 0) \quad \left. \vphantom{\begin{matrix} f(0) = \frac{4}{-8} = (-\frac{1}{2}) \\ (0, -\frac{1}{2}) \end{matrix}} \right\} \checkmark$$

$$x=2 \text{ and } x=(-2)$$

$$f(x) = \frac{\frac{1}{2}x + \frac{4}{2}}{2 - \frac{8}{2}x^2}$$

$$\text{as } x \rightarrow \pm\infty \quad f(x) \rightarrow 0$$

$$\therefore y=0 \quad \left. \vphantom{\begin{matrix} f(x) = \frac{\frac{1}{2}x + \frac{4}{2}}{2 - \frac{8}{2}x^2} \\ \text{as } x \rightarrow \pm\infty \quad f(x) \rightarrow 0 \end{matrix}} \right\} \checkmark$$

(iii) $f(x) = \frac{x+4}{2x^2-8}$ Quotient Rule $u = x+4$ $v = 2x^2-8$
 $f'(x) = \frac{(2x^2-8) - 4x(x+4)}{(2x^2-8)^2}$ ✓ $\frac{du}{dx} = 1$ $\frac{dv}{dx} = 4x$

$$= \frac{-8-2x^2-16x}{(2x^2-8)^2}$$

$$f'(x) = \frac{-8-2x^2-16x}{4(x^2-4)^2}$$

$$= \frac{-(4+8x+x^2)}{2(x^2-4)^2}$$

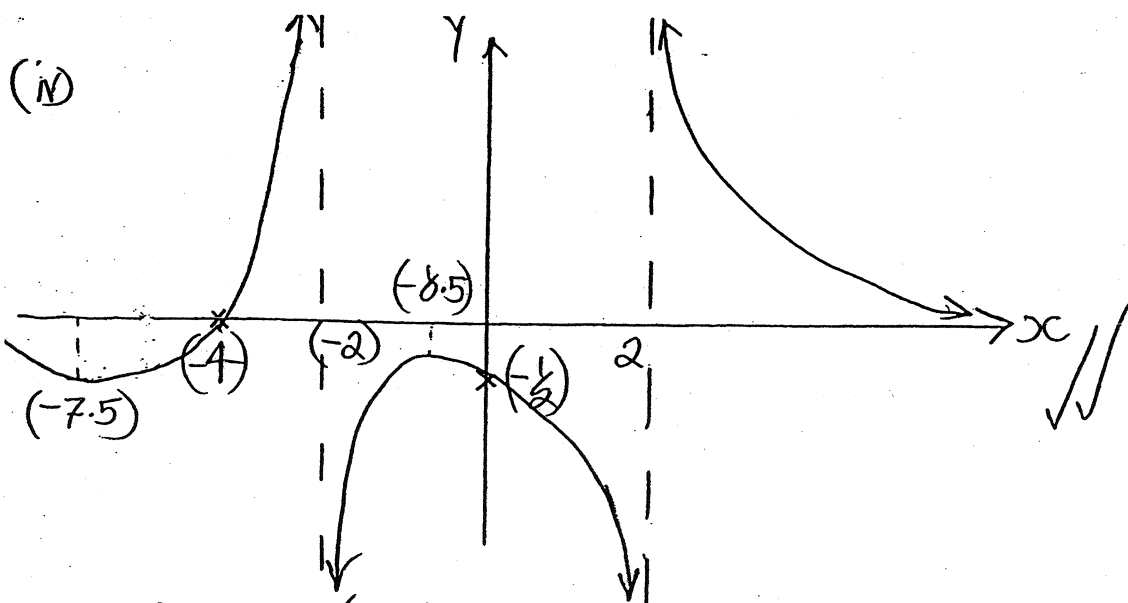
Horizontal Gradient $f'(x) = 0 \Rightarrow x^2 + 8x + 4 = 0$

$$\Delta = 8^2 - 4 \times 4$$

$$= 64 - 16$$

$$= 48$$

Zero Gradient at $x = -4 \pm 2\sqrt{3} \approx -0.5$ $x = -4 \pm 2\sqrt{3}$ ✓
 and $x = -4 - 2\sqrt{3} \approx -7.5$ $= -4 \pm 2\sqrt{3}$



$$f(x) = \frac{(x+4)}{2(x+2)(x-2)}$$

as $x \rightarrow 2^-$ $f(x) \rightarrow \frac{6}{2 \times 4 \times (-\epsilon)} \rightarrow -\infty$ (Not Req'd)

as $x \rightarrow 2^+$ $f(x) \rightarrow \frac{6}{2 \times 4 \times \epsilon} \rightarrow \infty$

as $x \rightarrow (-2)^-$ $f(x) \rightarrow \frac{2}{2 \times (-4) \times (-\epsilon)} \rightarrow \infty$

$x \rightarrow (-2)^+$ $f(x) \rightarrow \frac{2}{2 \times (-4) \times \epsilon} \rightarrow -\infty$

12